GOW30 Tool: A Collaborative Approach to Road Design Decisions

Final Report

Amsterdam Living Laboratory

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Executive Summary

This report presents the outcomes of a Living Lab research project focused on improving decision-making around the redesign and implementation of GOW30 roads in Amsterdam. Following the city-wide introduction of a 30 km/h speed limit on arterial roads, the project addresses a key challenge: how policy ambitions and technical guidelines translate into credible, context-sensitive road designs that support speed compliance, safety, and liveability.

Using a Living Lab approach, the research combines policy review, expert interviews, and a co-creation session with expert stakeholders from government, consultancy, research, and advocacy organisations. The findings show that GOW30 implementation is not primarily constrained by a lack of technical knowledge, but by fragmented decision-making structures, limited shared understanding, and difficulties in communicating design trade-offs across disciplines and stakeholder groups.

Based on these insights, the project develops a prototype digital decision-making support tool aligned with the mission of the DRO–DMI consortium to foster innovative, accessible, and liveable mobility systems. The tool translates established GOW30 design criteria and contextual factors into a structured framework that enables users to explore and visualise road-specific design scenarios. Rather than prescribing optimal solutions, it supports transparent comparison of options and facilitates structured dialogue between expert and non-expert stakeholders, including residents, transport operators, and emergency services.

The report concludes that such tools can strengthen public dialogue and governance by making complex design decisions more explicit, comparable, and accessible. It demonstrates the potential of combining Living Lab research with digital visualisation using AI-models to support collaborative decision-making. Further development could enhance its applicability within Amsterdam and its transferability to other urban contexts facing similar challenges.

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1. Background

1.1 Introduction

On 8 December 2023, the Municipality of Amsterdam implemented one of its most radical mobility shifts in recent decades: the reduction of the speed limit from 50 km/h to 30 km/h on a total of 270 kilometres of the city's arterial roads (Municipality of Amsterdam, 2024). These roads are unique in the sense that they carry a dual function: a traffic flow function where motor traffic including e.g. buses should be able to flow freely, while also serving as residential or mixed-use roads with homes, shops, and daily urban life directly along their edges. Nationally, this road type is legislatively categorised since 2021 as a gebiedsontsluitingsweg 30 km/h or GOW30 in short (CROW, 2021b). By lowering the speed limit, the city aims to strengthen road safety and improve liveability, particularly for those who live, work, cycle, or walk adjacent to these roads.

The transition from GOW50 to GOW30 does not stand alone. It forms part of a broader tradition of mobility reforms in Amsterdam, prioritising safe, accessible, and human-centred roads. Within this wider policy agenda, the challenge is not only technical, i.e. how to manage traffic flow efficiently at lower speeds, but organisational and cultural: how to redesign arterial roads so they look and feel like 30 km/h environments, and how to support behavioural adaptation from residents and road users.

Recent evaluation of the policy in a published report by the Municipality of Amsterdam (2025) further underlines the relevance of this topic. The analysis shows significant improvements in perceived and measured road safety, reductions in noise levels, and generally positive liveability outcomes. Importantly, concerns regarding reduced accessibility or worsening congestion have not materialised to the degree initially feared. With the first effects now strongly in place, the question shifts from why Amsterdam implemented GOW30 to how these roads should be redesigned to true low-speed environments, and how the decision-making process behind such redesigns can be improved.

This project relates to ambitions of the DRO-DMI consortium, a collaborative initiative by Amsterdam Institute for Advanced Metropolitan Solutions (AMS), Municipality of Amsterdam, mobility consultancy firm Goudappel, and other partners. Their shared mission is to develop innovative digital tools that help cities remain accessible and liveable amid rising demands on mobility networks (DRO-DMI, n.d.). Within this framework, a Living Lab approach provided by AMS Institute and our relating master's program Metropolitan Analysis, Design & Engineering (MSc MADE) offers applied research setting in which academic research, real-world experimentation, and stakeholder engagement intersect (Dijkstra & Joore, 2025).

1.2 Context and problem statement

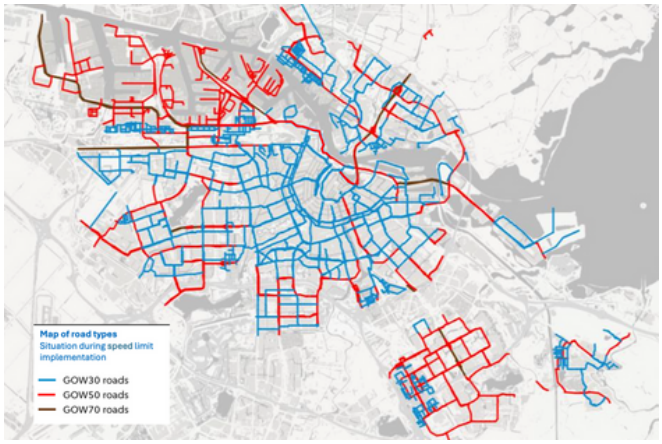


Figure 1: The 2023 GOW30 policy change, mapping where all roads in blue were lowered in speed from 50 km/h to 30 km/h (Municipality of Amsterdam, 2025).

The figure above illustrates the spatial distribution of Amsterdam's 2023 speed-limit transition, showing all former 50 km/h roads (GOW50) that were converted to GOW30. This shift addressed concerns about traffic safety and road liveability along these arterials, where urban and living functions could not be safely accommodated anymore with a maximum speed of 50 km/h. While early evaluations indicate that the safety and liveability objectives of the policy have largely been met, the transition has also given rise to new and other interlinked challenges requiring further attention.

To begin with, a key challenge emerging from the transition relates to its public acceptance and perceived legitimacy. Even though the policy has produced measurable improvements, questions persist about whether the measure restricts motor-vehicle accessibility or signals an unintended reduction of car use in the city. These concerns highlight a need for ways that can clearly communicate how the wider social and environmental benefits of GOW30 outweigh marginal losses in travel speed and efficiency (Municipality of Amsterdam, 2025).

A second challenge arises from a misalignment between road design and the intended speed behaviour. Many roads designated now as GOW30 still resemble their former 50 km/h layout, lacking visual design cues that intuitively support slower driving. This discrepancy negatively affects the credibility of the speed limit, as the physical environment communicates a higher speed than the legally posted limit implies. Consequently, the road fails to intuitively guide driver behaviour, creating a disconnect between the rule and the reality (Gargoum et al., 2016). This mismatch highlights the importance of examining where and how design interventions are required to create credible and self-explaining 30 km/h roads.

Another challenge concerns the complex decision-making environment around GOW30 redesign. The process involves a broad set of stakeholders: e.g. municipal departments, consultants, public transport companies, and civic groups, all of which have different interests. Without a structured guidelines for evaluating design options or weighing competing interests, design choices risk becoming opaque, inconsistent, or overly reliant on professional intuition. Addressing this requires a more transparent and collaborative approach in which decisions can be systematically explored and compared by all stakeholders involved.

1.3 Research scope, aim, and questions

Current decision-making processes affecting the design of and resulting speed limit compliance on GOW30 roads remain under-researched. The aim of this research is to explore the effect of utilising a public dialogue approach and decision-making support tool to make the decision-making process for road redesigns of GOW30 roads in Amsterdam more transparent and collaborative between key stakeholders. The value of public dialogue lies in the practical application of the Living Lab method.

Following this, the central research question to this project is:

How can a decision-making support tool contribute to the public dialogue on the governance and implementation of GOW30 in Amsterdam?

To support this main question, four sub-questions are formulated to guide the research process step-by-step. The aim of the first sub-question is to understand the current decision-making environment: *“What are the key decision-making structures and challenges shaping the implementation of GOW30 road design in Amsterdam?”*

Next, a structured design framework is constructed utilizing policy reviews and insights from experts in the field: *“How can existing design criteria, expert reasoning, and policy guidelines around GOW30 be formalised into a transparent and structured design framework?”*

Consequently, stakeholder perspectives within the decision-making framework are explored during a co-creation session: *“How do expert stakeholders perceive the challenges and opportunities within the current GOW30 decision-making and implementation process?”*

Finally, design and decision-making insights are integrated into a (prototype) product with the general aim to improve governance and design practice of GOW30 implementation: “How can insights from the Living Lab process be integrated into a decision-making support tool to enhance its effectiveness of GOW30 implementation?”

Final product

The goal of our eventual product, developed in line with the collaborative commitments formulated with DRO-DMI, is to create a digital tool that enables more effective public dialogue and fosters open collaboration, particularly between experts and non-experts in the field of transport. Through this tool, we aim to strengthen shared understanding and support constructive engagement between (municipal) stakeholders and involved residents.

Project scope

The scope of this project is clearly defined by the constraints of the Living Lab course in the MADE master’s program. First, the timeframe for this project is restricted to four months, spanning from September 2025 to December 2025. Second, the project is initiated by DRO-DMI with the shared mission to gain knowledge on how to innovatively orchestrate the use of public space. Therefore, our findings are aimed to support these efforts. Lastly, we focus on GOW30 roads in only Amsterdam. While GOW30 has been implemented in multiple Dutch cities, Amsterdam offers the most practical and analytically coherent case study for our purposes. We reside and work in the city, and our research group maintains extensive relationships with local stakeholders, which facilitates access to relevant expertise, data, and contextual knowledge.

1.4 Reading guide

This report follows our step-by-step process during the Living Lab project. The description of the process contains background theory for context and better understanding of current decision-making processes in the study area. Subsequently, the living lab approach and methods used in this project are explained. After the process, the results are presented in order of the research questions. The insights from our literature review, interviews and co creation are incorporated in this chapter. Afterwards, the findings are synthesized. In the final chapter we elaborate on our research outcomes and final digital tool, where its specifications and limitations are also discussed.

2. Process

This chapter outlines the research process used to explore GOW30 (re)design and decision-making in Amsterdam. It explains the theoretical grounding, the methods applied, and the sequential phases of our Living Lab approach, showing how theory, expert insights, and policy analysis collectively shaped our understanding of the relevant challenges and opportunities.

2.1 Literature review

This literature review covers three key theoretical fields: driver behaviour and road design, explaining how environment and compliance interrelate; the GOW30 policy framework in Amsterdam, outlining current governance and implementation practices; and decision-making theory and support tools, providing the foundation for developing a structured tool that enhances dialogue and decision-making on GOW30 design.

2.1.1 Designing roads to steer compliance

Previous research on driver speed compliance demonstrates that traffic behaviour arises from a continuous interaction between cognitive processes, personal characteristics, and the surrounding road environment. Yadav & Velaga (2021) show that compliance is strongly context-dependent, with drivers adapting more effectively to lower speeds in urban settings than in rural ones. Their findings highlight that the perceived credibility of a speed limit depends on how intuitively the environment communicates it, pointing to the psychological and cognitive foundations of speed choice. Gargoum et al. (2016) extend this cognitive-environmental relationship by examining how specific physical road features influence compliance. While Yadav & Velaga demonstrate that environment broadly affects how drivers interpret and adapt to speed limits, Gargoum et al. quantify the influence of concrete design elements such as lane widths, roadside friction, and visual enclosure. Their work shows that compliance increases when the physical form of the road aligns with the intended limit. This positions road characteristics not merely as a contextual backdrop but as a primary behavioural determinant that can actively steer drivers toward compliance.

Orsi et al. (2025) build on these behavioural and environmental insights by introducing a computational method that operationalises them. Using machine-learning models trained on road-view imagery, they quantify how visible design characteristics affect expected compliance. Their research synthesizes the previous concepts into a predictive framework capable of estimating driver compliance from the visual attributes of a road. Together, these studies present a clear conceptual baseline: cognitive tendencies shape behaviour, environmental cues modulate those tendencies, and emerging computational tools can now measure and predict this relationship at larger scales.

Efforts to improve speed limit compliance in urban environments tend to follow two complementary design theories: self-explaining roads and traffic calming measures. Both aim to reduce driver speed, yet they operate through different yet intertwining behavioural mechanisms.

Self-explaining roads (SERs) are based on the idea that a road's physical form can instinctively guide safe driving behaviour. Geem et al. (2013) define SERs as road environments that elicit appropriate behaviour through design alone. The concept rests on two psychological processes-categorisation and expectancy. Over time, drivers form "prototypical representations" of road types; when a road's appearance is consistent and clearly distinguishable from other categories, predictable and safe behaviour follows naturally. Qin et al. (2020) strengthen this claim empirically by quantifying how characteristics in SERs shape speeding behaviour, demonstrating that design coherence is a key predictor of speed-limit compliance.

When self-explaining conditions cannot be fully achieved, which happens particularly on arterial roads with mixed functions, traffic calming measures (TCMs) become necessary. TCMs encompass physical or visual interventions explicitly intended to reduce vehicle speeds, such as speed bumps, chicanes, narrowed lanes, raised crossings, or gateways. Distefano & Leonardi (2022) show that TCMs create a fluctuating speed profile that forces drivers to slow down at strategic points, compensating for road environments that do not inherently induce compliant behaviour. Pirdavani et al. (2025) further highlight the usefulness of TCMs in high-to-low speed transition zones, suggesting that well-designed gateway elements can reduce reliance on costly enforcement and support the broader goal of creating self-explanatory networks.

TERs and TCMs are inherently connected and share a lot of overlap in execution, however they are key in understanding the mechanisms behind speed limit compliance. Together they form the design foundation for understanding how GOW30 roads can become both legible and credible to drivers.

2.1.2 GOW30 road-category

The contemporary principles of GOW30 are largely explained through the Sustainable Safety framework articulated by the Institute for Road Safety Research (SWOV, 2019). Historically, GOW30 emerged from a political and professional agenda to resolve longstanding inconsistencies between road function and road design within urban areas. SWOV identifies that many urban distributor roads had evolved into ‘grey roads’: roads that simultaneously carried characteristics of verblijfsfuncties (residential functions), typically associated with erftoegangswegen 30 km/u (ETW30), and ontsluitingsfuncties (through- or distribution functions), associated with gebiedsontsluitingswegen 50 km/u (GOW50).

Because these roads served both as places where people live, walk, and cycle, and as corridors for motor-vehicle movement, they no longer fitted neatly into either category.

This mismatch created situations where the functional role of the road conflicted with its design and user composition, producing safety risks, higher speeds, and lower liveability. The introduction of the GOW30 category was therefore intended to remove this ambiguity by offering a design and regulatory framework that could better balance road capacity and the societal goals of safety and liveability within built-up urban contexts (CROW, 2021b; Municipality of Amsterdam, 2024; SWOV, 2019).

Building on these conceptual foundations, Schepers et al. (2025) analyse how specific combinations of design elements within the Dutch built environment shape driving speed on 30 km/h roads. Their findings show that speed behaviour cannot be attributed to isolated features such as lane width or the presence of cyclists; instead, it emerges from interactions between road geometry, frontage activity, and spatial enclosure. This empirically reinforces SWOV’s assertion that recognisable and coherent design is essential to produce credible speed regimes, and it demonstrates that context-dependent effects must be considered to understand actual driver responses.

The report by Municipality of Amsterdam (2025) localises the initial outcomes of the GOW30 principles by documenting how the city-wide transition has thus far improved safety and liveability. The report notes reductions in crash rates, noise levels, and perceived traffic pressure, highlighting several roads where lower speeds have directly enabled more space for pedestrians and cyclists. These effects intersect with a broader cultural shift in Amsterdam, where the 30 km/h limit is becoming the normative urban standard and where car use is increasingly deprioritised. Compared with other Dutch municipalities, Amsterdam adopts a more assertive interpretation of the national mandate, raising questions about how local political ambition and spatial conditions shape differentiated municipal strategies toward GOW30 implementation.

2.1.3 Design process: decision-making tools & public dialogue

Aside from the technical considerations that need to be studied when re-designing a road, one may also look at the way in which these decisions on interventions are made. There are multiple theoretical understandings which guide this process. Decision-making in traffic management has traditionally been dominated by top-down, expert-driven planning approaches. However, Immers & van der Knaap (2011) argue for a shift toward a living systems approach, in which traffic systems are viewed as dynamic and adaptive rather than static or centrally controlled. Within such a system, each stakeholder, whether it is a policymaker, designer, or user, plays an integral role in maintaining balance and resilience. This argument fits within the broader governance principles outlined by Tasan-Kok (2021), who describes urban governance as inherently multi-actor, involving the interplay of institutional, professional and civic stakeholders whose interests and power relations must be managed rather than simplified. When applied to GOW30, this theoretical framing suggests that road redesigns should not solely reflect technical criteria but also respond to social and institutional dynamics across various scales.

Building on this systems perspective, Bertolini & Brömmelstroet (2011) explore how improved **decision-making (support) tools** can help structure the complexity of urban transport planning. They advocate for tools that bridge the gap between analytical modelling and communicative planning, allowing stakeholders to engage meaningfully in decision processes without reducing problems to purely quantitative terms. Such tools, when used in conjunction with public dialogue, enable a two-way exchange between data and lived experience, balancing technical insight with local knowledge. In the GOW30 context, this approach can guide how policymakers and consultants evaluate trade-offs between safety, accessibility, and liveability in arterial road redesigns.

Finally, Aspen (2023) demonstrates in particular how decision-making (support) tools can operationalize this integration by translating data-led insights into participatory scenarios. Rather than replacing human reasoning, these tools facilitate transparency and collective learning among stakeholders. By modelling different spatial or behavioural outcomes, they support more informed and balanced governance decisions. Within the Amsterdam GOW30 transition, such approaches may help structure stakeholder interactions, clarify design priorities, and create a shared language between engineers, policymakers, and urban designers. Ultimately strengthening both the governance and implementation of low-speed road designs.

2.1.4 Conceptual framework

The conceptual diagram synthesizes the theoretical foundations discussed by tracing how insights from cognitive driver behaviour, road design theory, and decision-making approaches collectively shape the implementation of GOW30 in Amsterdam. The starting point here is the recognition that driver behaviour is context-dependent: both cognitive tendencies and environmental cues determine whether drivers comply with a lower speed limit, as demonstrated in behavioural studies. This behavioural core directly feeds into two major strands of design theory: self-explaining roads and traffic-calming measures. Both strategies translate behavioural insights into design principles and together form the theoretical backbone of the GOW30 design framework shown in the diagram.

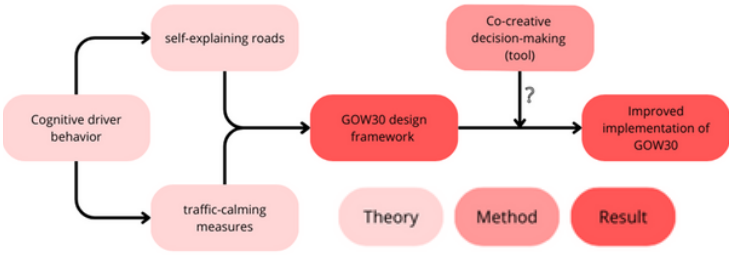


Figure 2: the conceptual diagram for this research.

The GOW30 design framework represents the methodological centre of the project: it formalises how behavioural knowledge and environmental factors can be translated into (categorised) criteria for evaluating and selecting appropriate GOW30 interventions. This aligns with the broader theory on decision-making tools in transportation planning, which emphasises structure and transparency in complex urban design processes. However, theory also highlights that improved decision-making does not rely on traditional (design) planning alone. Implementation requires navigating multiple institutional actors, perspectives and priorities.

The real knowledge gap here lies in the practicalities of achieving this effectively through active stakeholder engagement. For that reason, the diagram incorporates co-creative decision-making as an essential methodological complement to the design framework. Co-creation connects technical reasoning with experiential and professional knowledge from different stakeholders, helping to bridge gaps between behavioural theory, institutional constraints, and practical implementation. The precise interaction between co-creative decision-making and the GOW30 design framework is the key analytical relation to this research (as represented by the question mark indicating this is also the knowledge gap), In the end, this interrelation converges towards the same outcome: improved implementation of GOW30. The diagram thus illustrates how theoretical insights on behaviour, design strategies, and decision-making processes interact to support a more credible, effective, and context-sensitive transition to 30 km/h roads in Amsterdam.

2.2 Methods

This section outlines the methodological approach underpinning our study. We first describe the Living Lab methodology and clarify its relevance to the research topic and guiding questions. We then detail our data collection strategy, which consists of expert interviews, policy analysis, and a co-creation session. Finally, we explain the analytical procedures applied to these data to derive our findings.

2.2.1 Living Lab workflow

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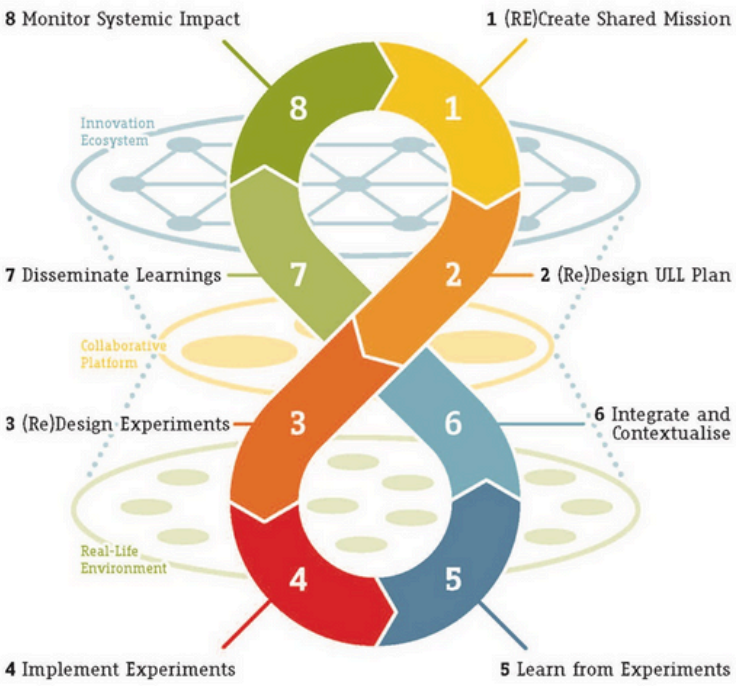


Figure 3: the Urban Living Lab cycle (Dijkstra & Joore, 2025).

Our research followed a Living Lab method designed to iteratively explore, test, and refine insights about the implementation and redesign challenges of Amsterdam's GOW30 distributor roads. Visualized in Figure 3, Living Labs are collaborative, real-world research environments in which stakeholders jointly and iteratively develop, test, and evaluate interventions under close-to-reality conditions (Dijkstra & Joore, 2025). It is an infinite loop of iterative steps creating, experimenting, and disseminating on three levels. First the innovation system, which is the layer where innovation ideas are created and monitored. Then comes the collaborative platform which is where collaboration is the main environment. In this level, a plan is developed as well as design experiment and insights from the experiments are integrated and contextualised and finally disseminated from this level to the innovation ecosystem. The last level is the real-life environment, where experiments are executed. After the execution of the experiment in the real-life environment, the insights and are integrated and contextualised on the collaborative level and finally disseminated and impact monitored back in the innovation ecosystem. Following Dijkstra & Joore (2025), our Living Lab combines co-creation and iterative learning, while adopting a hybrid top-down/bottom-up research structure. The municipal policy documents are the foundation of our framework, whereas in the co-creation we focused on consultants and researchers to collaboratively explore what the GOW30 transition requires in practice.

The applied workflow, as visualised in Figure 4, of this project contains 3 phases. The first phase starts with gaining more background knowledge on the topic and finding a shared mission which relates to the first steps of the Living Lab cycle. Then we started exploring and experimenting with our tool and collaborating with stakeholders during a co-creation session which aligns with step 3 until 5. Finally, we synthesize our findings in this research paper and disseminate our learnings and tool-product via our Interactive Living Lab website.

Throughout these phases, the Living Lab structure allowed for continuous iteration: early findings guided subsequent data collection, stakeholder inputs refined research questions, and expert insights informed our evolving conceptualisation of what a GOW30 distributor road should be as well as iterate on the tool's logic and user experience. This workflow ensured methodological rigor while maintaining the flexibility needed to study a complex, multi-actor urban transition in real time.

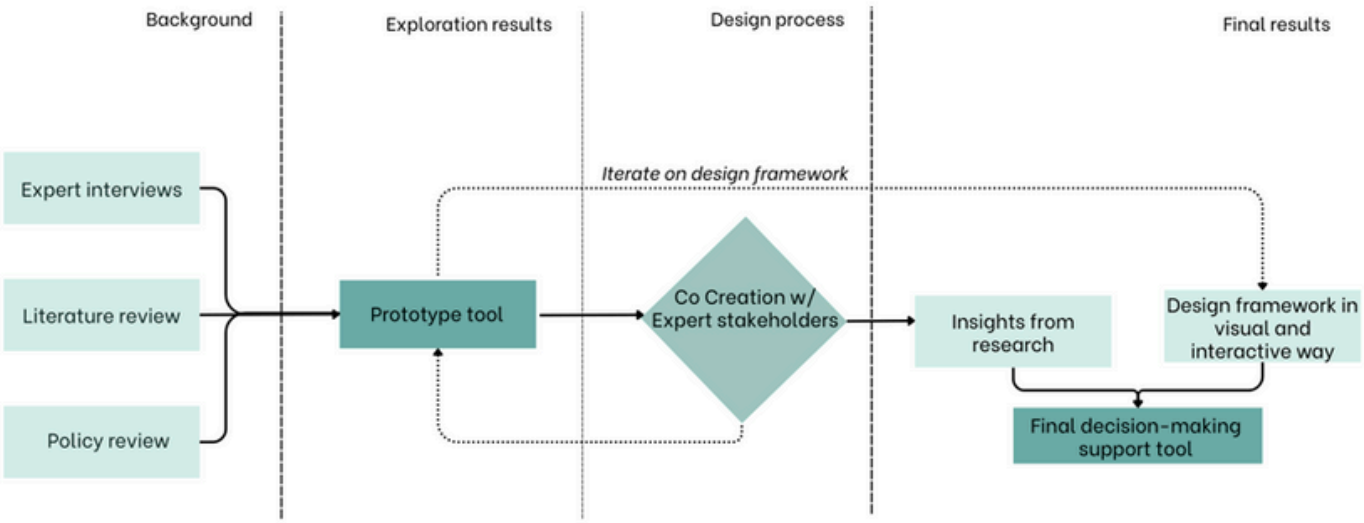


Figure 4: the workflow diagram for this research project.

2.2.2 Policy review

To facilitate a more transparent and systematic approach to the redesign process of GOW30 roads, a review of current policies was conducted to identify the governing policies, and the recommended design elements that specify how a GOW30 road should ideally be configured.

A selection of policy documents was made based on their relevance to the redesign process of GOW30 roads in Amsterdam specifically. One of the key documents that guide GOW30 road design is *'Handreiking voorlopige inrichtingskenmerken GOW30'*, a document published by CROW (2023). CROW is a Dutch knowledge institute that develops and publishes advisory standards and guidelines for infrastructure and public space that are widely used by road authorities and municipalities. This specific document stipulates preliminary recommended design elements for GOW30 roads, such as pavement type, road widths, and bicycle facilities.

Additionally, *'Handboek Rood'*, published by the Municipality of Amsterdam (2024), provides standards for road designs in Amsterdam and proposes a number of categorical road layouts. Finally, a few other documents were reviewed, including *'Gereedschapskist GOW30'*, published by the traffic and infrastructure consultancy Goudappel (2025). This document also proposes various potential GOW30 road layouts and includes practical recommendations on which design is most appropriate given certain criteria.

The insights gained from this review of policy documents served as the technical foundation for developing the decision-making support tool. The design elements and recommended road layouts identified in the documents informed the structure of the tool, to ensure the tool reflects existing standards. The operationalisation and logic behind the tool will be discussed in further detail in §3.1.

2.2.3 Stakeholders

The implementation and future redesign of Amsterdam's GOW30 distributor roads involve a diverse and interdependent network of stakeholders spanning different domains. Understanding this landscape is essential for the Living Lab, as the transition to 30 km/h requires coordinated action across policymaking, road design, and technical expertise. To map this environment, we created a power-interest matrix (see Figure 5) to analyse each actor's influence on decision-making and interest in the topic.

First, we listed all relevant stakeholders relating to our topic. Starting with our case initiator, DRO-DMI, we mapped out the Municipality of Amsterdam, AMS institute and Goudappel. Besides Goudappel, consultancy firm DTV have also published extensive research on road design and redesign policies surrounding GOW30. Furthermore, other knowledge institutes that are involved in Amsterdam GOW30 policy and road design are TU Delft, SWOV, and CROW. Finally, civil parties and advocacy groups are related stakeholders as well.

Second, we analysed their power and interest in GOW30 implementation processes. The civil and advocacy stakeholders hold the most interest as they are the most vulnerable road users for whom speed limit compliance of car users is extra important. Then, the knowledge stakeholders have on average slightly more power than civil stakeholders however, some are more specialised in this topic than others. Thus, those with more specialized knowledge are placed with a higher interest on the matrix. The Municipality of Amsterdam has the highest power given that they implement the policy and hold the executive power on this topic. However, the municipality also must deal with all other governmental tasks besides tackling challenges on GOW30 roads, hence their interest is not as high.

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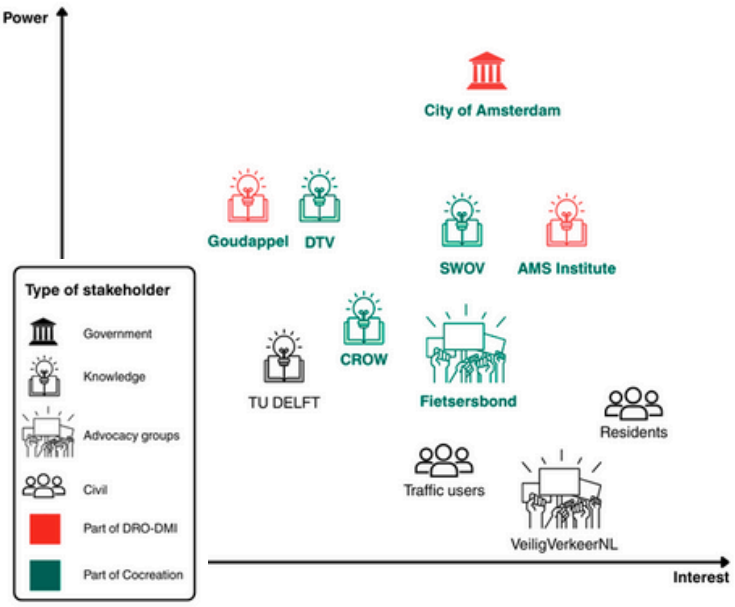


Figure 5: the power-interest matrix of the stakeholders in this research.

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2.2.4 Expert interviews

To gain deeper insight into the design, behavioural, and institutional challenges underlying Amsterdam's GOW30 transition, we conducted a series of semi-structured expert interviews. This method was selected because it enables the exploration of nuanced, context-dependent knowledge that cannot be captured through quantitative datasets alone. Open-ended dialogue allowed experts to articulate their interpretations of policy decisions, design constraints, organisational responsibilities, and stakeholder tensions.

Sample and participants

In total, ten expert interviews were conducted with experts directly involved in, or closely connected to, the GOW30 policy. The interviewees represented a diverse range of institutions, ensuring a cross-disciplinary and multi-stakeholder view of the transition.

Interviewees	Role	Organisation
Person 1	Head of Traffic Design | GOW30 Toolbox	DTV Consultants
Person 2	Lead & Program Developer | Driving behaviour & road design research	MIT Senseable City Amsterdam | AMS Institute
Person 3	Researcher | Driving behaviour & road design research	MIT Senseable City Amsterdam | AMS Institute
Person 4 Person 5	Senior Project Manager Urban Design & Traffic Engineer	Municipality of Amsterdam | dep. of Traffic & Public Space | DRO Amsterdam | GOW30 Project Team
Person 6	Head of Traffic Design | GOW30 Toolbox	Goudappel
Person 7	Traffic Design Engineer Proof of Concept	DRO-DMI | AMS Institute
Person 8	Project Manager Mobility & Expert in Road Design	Knowledge Platform CROW
Person 9	Researcher on Road Safety	Institute for Road Safety Research (SWOV)
Person 10	GOW30 researcher on road safety	Institute for Road Safety Research (SWOV)

Figure 6: an overview of all interviews conducted for this project.

Each interviewee held a specialistic expertise, ranging from traffic design and road safety to programme management and policy strategy, allowing us to triangulate perspectives across technical, operational, and political domains. Interview durations ranged from 45 to 75 minutes and were conducted either in person or online depending on availability.

Interview Focus and Themes

A universal interview guide was developed around four overarching themes:

1. **Policy and implementation;** Decision-making structures, organisational roles, and barriers affecting the implementation of GOW30.
2. **Design and technical aspects;** Road and technical challenges and the role of design tools in creating self-explaining roads with a credible speed limit.
3. **Stakeholder collaboration;** Interactions between municipal departments, consultants, and research organisations, and the conflicts or shared interests within these relationships.
4. **Evaluation and future outlook;** Expected impacts; lessons for future redesign processes; and reflections on Amsterdam's unique mass-transition strategy.

This thematic structure allowed for consistency across interviews while leaving room for emergent topics, such as compliance enforcement, feasibility constraints, and political sensitivities.

Interview procedure and coding

At least two team members attended each interview: one as the primary interviewer and one as an observer and note-taker. With informed consent, all interviews were recorded using the Microsoft Teams transcription tool. The transcripts were analysed using an inductive thematic coding approach (Fereday & Muir-Cochrane, 2006). For this, two researchers independently coded the interviews, highlighting quotes and emerging categories. These were subsequently discussed, merged, and organised into thematic clusters that formed the basis for our analytical findings. This approach was adopted to ensure proper research validity and reliability in this part of the project.

Role in the Living Lab Workflow

The expert interviews served both exploratory and confirmatory functions within the overall Living Lab structure. Early interviews helped frame the core challenges in road design, behavioural compliance, and institutional alignment. Meanwhile, later interviews were used to validate or nuance patterns emerging from contextual analysis, stakeholder mapping, and municipal documentation. The cumulative insights informed the iterative refinement of our research focus and the development of criteria for redefining the identity of GOW30 roads.

By triangulating expert input with spatial analysis, policy documents, and stakeholder mapping, the interviews provided a robust empirical foundation for understanding how Amsterdam's GOW30 distributor roads are being interpreted, designed, and governed in practice.

2.2.5 Co-creation session

The key method utilised in this research was hosting a co-creation session aligned to the Living Lab handbook by Dijkstra & Joore (2025). Co-creation is the practice of collaborating with diverse set of stakeholders to facilitate meaningful participation strengthening the design process. Collaborative design efforts can help us understand the stakeholders' needs better and lead to unexpected ideas.

For this, we invited a focused group of stakeholders directly involved in the technical and governance aspects of GOW30 road design. This included our project partners from the DRO-DMI consortium (Municipality of Amsterdam, AMS Institute, and Goudappel), consultancy firm DTV, knowledge institutes SWOV and CROW, and advocacy group Fietsersbond. We deliberately limited the group to professional organizations to allow for deeper discussions on design logic, technical challenges, and decision-making structures. The co-creation session consisted of two main parts.

Design workshop

The first part of the session focused on collecting input from the stakeholders by explaining our logic behind the tool to the stakeholders as well as demonstrating the prototype on an example case of Pieter Calandlaan in Amsterdam. We used an interactive workshop format to actively engage the stakeholders. After a brief presentation from our side introducing the Living Lab method and demonstrating our tool, stakeholders were divided into small mixed teams of two to three people. Each team consisted of experts with diverse backgrounds to encourage diverse perspectives.

For the workshop, the teams worked through a structured worksheet (Appendix 2) consisting of 3 parts. First, the stakeholders listed and elaborated on criteria they consider most important when redesigning a road. Second, each team selected their four most important criteria and translated them into concrete design measures that could be applied to a GOW30 road. In the final part, teams applied these measures to the example case by drawing a simple cross-section of the Pieter Calandlaan, showcasing how their measures could change the road. To wrap up the workshop, each team briefly presented their design to the group.

Panel discussion

The second part of the session was centred on gaining insights into the institutional mechanisms behind the decision-making processes for GOW30 roads, as well as understanding behavioural change and cultural norms around speed-limit compliance. A panel discussion format was chosen as it enables semi-structured, interactive input from all participants, while also creating space for new perspectives, experiences and interpretations to emerge.

We began by identifying (potential) tension points from the literature, policy documents and expert interviews. For instance, the interview transcripts revealed recurring frictions concerning the technocratic nature of GOW30 road design, the fragmented stakeholder landscape, and persistent departmental silos. These and other observations were translated into concise "either-or" tension points, formulated deliberately to make underlying dilemmas explicit, tangible and debatable. As the invited experts preferred to conduct the co-creation in Dutch, the statements were originally drafted in that language but are translated into English for this report.

Due to the dual goals of this part of the session, all tension points were grouped into two overarching themes. Each theme consists of six statements (twelve in total; see Appendix 3). The first theme, Governance, Design and Institutional Dynamics, encompasses dilemmas such as top-down regulation versus co-created input, central standards versus local adaptation, and siloed departments versus integrated ownership. The second theme, Behavioural Change, Design and Social Norms, reflects expert insights on enforcement, self-explaining roads and cultural adaptation, expressed through tensions like enforcement versus design, individual compliance versus collective responsibility, and rapid policy roll-out versus gradual norm change.

During the session, participants collectively picked two statements per theme (four in total) through a voting mechanism. These selected statements were then used as structured prompts to critically examine how the interpretations drawn from interviews resonated with the broader group of stakeholders. Rather than introducing new issues, the tone of the discussion was designed to "mirror back" the key tensions identified earlier and explore how different actors positioned themselves when confronted with concrete GOW30 road challenges and the ways in which the emerging decision-support tool may help them.

3. Insights

This chapter presents the key insights derived from the policy review, expert interviews, and the co-creation session, synthesising the most relevant findings for the development of the GOW30 decision-support tool. It reconstructs policy-based knowledge into an initial design framework and complements this with practitioner perspectives and collaborative reflections that capture both technical and institutional dimensions of GOW30 implementation.

3.1 Design framework

To start, an initial technical framework is presented based on insights gathered from policy guidelines and design documents. This initial version served as the foundation for subsequent tool development based on the results of additional data sources (expert interviews, co-creation with experts).

3.1.1 Design variables

The first step that constitutes the underlying framework of the decision-making tool is to identify the tangible interventions that a design can vary in within a GOW30 road categorization: the possible design variables. Based on the reviewed documents presented in §2.2.4, the design elements (*inrichtingskenmerken*) and recommended road layouts were studied to determine which overarching design variables recur consistently across the various layouts (CROW, 2023; Goudappel, 2025; Municipality of Amsterdam, 2024).

Bicycle facilities

In the Netherlands, cycling infrastructure can generally be found in a number of distinct types. Firstly, on-road bicycle lanes (*fietsstroken*) are marked lanes on the carriageway itself, typically distinguished by red asphalt, white edge lines, and a painted bicycle symbol, signalling to traffic users that this space is primarily intended for cyclists (see Figure 7a). Secondly, fully separated bicycle lanes (*fietspaden*) are physically segregated from the main roadway, providing cyclists with their own dedicated space separated from motorised traffic (see Figure 7b).

CROW distinguishes these facilities primarily by the level of traffic intensity and the function of the road (CROW, 2021a, 2023). For standard arterial roads with a maximum speed of 50km/h, referred to as GOW50, it prescribes a baseline bicycle solution of fully separated bicycle lanes, or redirecting cyclists to parallel service roads (*ventwegen*), which commonly have a lower maximum speed and therefore more appropriate in safely accommodating cyclists. Due to the large difference in speed between motorists and cyclists, mixing these two kinds of traffic users is discouraged to ensure safety (CROW, 2023).

Service roads, which are usually categorised as ETW (*erftoegangswegen*), tend to have lower traffic volumes and lower average speeds as their primary function is to provide access to adjacent properties, such as homes and shops. Given these characteristics, mixing motorists and cyclists on the same road is the standard for ETW's. CROW advises against the use of on-road bicycle lanes, which aligns with the principle of shared space. This concept aims to enhance user awareness and safety through reduced formal regulation (Chou et al., 2025; CROW, 2023).

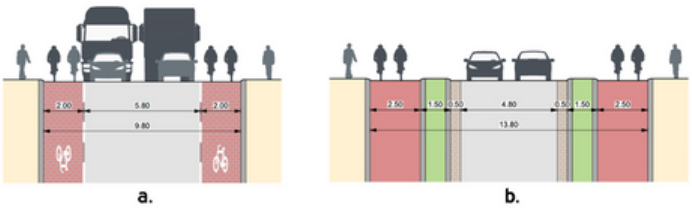


Figure 7: (a.) represents a road layout with fietsstroken, while (b.) represents a road with fietspaden (Goudappel, 2025).

Since GOW30 combines both the traffic flow function of a GOW50 and a staying function of a ETW30, CROW guidelines for bicycle facilities in a GOW30 recommend a solution that caters to both demands. Unlike ETW roads, CROW prescribes on-road bicycle lanes (*fietsstroken*) for GOW30 as the latter tends to see larger motor traffic volumes. In certain cases however, fully separated bicycle lanes (*fietspaden*) are also deemed appropriate. In its publication regarding GOW30 guidelines, CROW advises to implement fietsstroken on GOW30 roads and *fietspaden* on GOW50 roads, for the sake of recognisability.

Additionally, the Goudappel (2025) report indicates that the determining factor in which facility is optimal lies in the intensity of motorized traffic: if it exceeds 10.000 movements per day a *fietspad* is recommended. Additionally, the report suggests that available space acts as a constraint; a total road width of approximately 11,8 metres is required in order to apply a *fietspad*.

Road width

Available academic literature on factors influencing speed and driver behaviour indicates that certain road characteristics have a possible correlation with driving speed, one of which is road width. Drivers tend to instinctively drive faster on roads that are wider; these wide and open roads 'invite' them to drive at higher speeds regardless of the legal limit (Godley et al., 2004; Orsi et al., 2025; Schepers et al., 2025; SWOV, 2012).

In the analysis of recommended GOW30 designs, the width of the central carriageway, which is also commonly referred to as *middenrijloper* in Dutch, emerged as a common recurrent variable across the designs (Goudappel, 2025; Municipality of Amsterdam, 2024). The *middenrijloper* is defined as the section of the road situated in-between the bicycle facilities, such as *fietspaden* or *fietsstroken*.

This presents a trade-off in the design framework. On one hand, the central carriageway can be kept wider to accommodate traffic flow and heavy or large vehicles such as buses and emergency services. On the other hand, narrowing this section which helps stimulate lower speeds and simultaneously frees up space for e.g. wider bicycle facilities, or pavement space for pedestrians.

Paving type

In the Netherlands, roads are commonly paved with smooth asphalt, or with element paving, specifically bricks or clinkers (*klinkers*). The latter type is particularly common on residential service roads (ETW) that serve slow traffic in low volumes. The slowing effect of element paving on traffic occurs due to the increased tyre noise and vehicle vibration (Andriesse et al., n.d.; Loprencipe et al., 2019).

In its GOW30 guideline publication, CROW prescribes the use of element paving for GOW30 roads, aimed to create a 'credible speed limit', which is in line with the concept of self-explaining roads (Geem et al., 2013). It however also acknowledges that in numerous cases, paving the entire road with bricks or clinkers may not be feasible due to certain constraints. Examples are increased noise and vibration for residents, the alternative needs of buses and trams, and reduced ride quality and comfort for cyclists. In such cases, where asphalt is consequently preferred, CROW guidelines advise against applying it across the entire width of the road as this reduces recognisability and makes GOW30 hard to distinguish from a GOW50 (CROW, 2023)

Traffic calming measures

The final design variable identified through the policy review is Traffic Calming Measures (TCMs). Within this framework, these are operationalised as physical interventions explicitly designed to directly force a reduction in driving speed, such as speed humps (*drempels*), speed tables (*plateaus*), or chicanes. In the literature studied in §2.2.4, these were commonly referred to as speed reducers (*snelheidsremmers*).

3.1.2 Contextual criteria

In addition to the design variables, which represent the elements of a road that can be configured, the policy review also revealed a set of contextual criteria that influence the applicability of specific GOW30 design choices. These criteria emerged consistently across multiple policy documents (CROW, 2023; Goudappel, 2025; Municipality of Amsterdam, 2024). They act as parameters used to weigh trade-offs and which specific choice within a design variable is more suitable for a given road segment.

The following contextual criteria were identified in this initial phase:

1. *Land use*
2. *Pedestrian crossing needs*
3. *Pedestrian traffic intensity*
4. *Vulnerable groups (presence of schools, elderly homes nearby)*
5. *Public transport and emergency service routes*
6. *Freight/heavy traffic intensity*
7. *Motor vehicle intensity*
8. *Bicycle traffic intensity*
9. *Current compliance with posted speed limit*
10. *Parking intensity*
11. *Current road width*

It is important to highlight that these 11 criteria should not be treated with static or uniform importance; the degree of how each criterion is weighed is dependent on user input. The relative weight of each criterion is context-dependent, as in certain scenarios, the priority may be protecting vulnerable road users, while in another, the focus is optimal access for emergency services.

Furthermore, what is deemed as 'important' is inherently subjective and dependent on the user's perspective. Thus, in the design of the framework, pre-determined fixed weights were deliberately avoided, to avoid bias and turning the framework into a black box, where the logic is opaque to the end user.

It instead relies on a neutral baseline in which all criteria are assigned with an equal weight, which can then be manipulated by the user, by allowing them to select up to 3 criteria they deem as more important.

3.2 Expert interviews insights

In this section the main relevant insights from the ten expert interviews, both on design as well as decision-making processes, are presented.

Technical specificity of GOW30: bespoke design and limits of standardisation

Across the interviews, experts stressed that GOW30 cannot be treated as a uniform technical package. Instead, its implementation requires road-by-road customisation in which function, spatial constraints, and local context determine the appropriate combination of measures. As one designer from the Municipality of Amsterdam remarked, *"every road is different, so you have to make different choices, and at the moment a lot of that still happens by gut feeling."*

This bespoke approach is strongly tied to the hybrid character of Amsterdam's distributor roads. Many designated GOW30 roads still resemble their former 50 km/h layout, which, as a SWOV researcher noted, places *"recognisability really at stake"* and helps explain low speed compliance in several locations. Experts agreed that the challenge is not only identifying measures, such as narrowing lanes, adding humps, creating visual friction, but understanding which interventions fit which contexts.

Consultants emphasised that this knowledge base is still developing. Goudappel acknowledges that the transition is *"an ongoing learning process,"* with practitioners still determining what works where. Existing toolboxes and checklists were described as helpful but insufficiently specific; they provide possible measures without clarifying when each is appropriate. An AMS Institute researcher summarised this gap: *"there is a list of measures, but not a clear sense of when to apply which in which context."*

Multiple interviews also highlighted practical constraints that shape design decisions. Municipalities often synchronise road redesign with broader interventions like sewer reconstruction to limit costs and disruption.

While operationally logical, this sequencing means that the optimal redesign moment is frequently dictated by maintenance cycles rather than by the policy's technical requirements.

Stakeholder relations and institutional tensions

The interviews generally portray GOW30 as embedded in a dense and sometimes fragmented network of stakeholders. Interviewees repeatedly emphasised how many different groups are affected by the 30 km/h transition, each bringing their own priorities, tools, and vocabularies to the table. As one expert from Municipality of Amsterdam summarised, *"you have taxi branch lobbyists, the cycling organisations, neighbourhood groups, the public-transport operators... everyone has a different opinion."* This diversity creates a rich but complex decision-making environment in which, as the same interviewee noted, *"these groups cannot easily use each other's tools"*, reinforcing parallel rather than shared problem framings.

Beyond differences between stakeholder groups, the interviews also revealed strong disciplinary silos within the institutional landscape itself. Several experts described how traffic engineers, planners, policymakers, and enforcement agencies still tend to *"sit in their own world,"* making it difficult to coordinate design, policy, and behavioural measures for specific roads. One respondent argued that *"the different disciplines... traffic engineers, planners, policymakers, police... need to come much closer together,"* explicitly identifying interdepartmental fragmentation as a barrier to effective GOW30 implementation.

These institutional dynamics also shape how the municipal government interacts with residents. While many broadly support the ambition of safer, slower roads, their everyday experience often reflects the same gaps between policy, design, and enforcement noted by professionals. As Municipality of Amsterdam explained, residents frequently ask, *"Can my road also become 30?"* followed immediately by concerns that *"everyone is still driving too fast."*

This feedback illustrates how unclear responsibilities and inconsistent implementation can translate into frustration or perceived inaction. Rather than outright distrust, the responses primarily reveal a mismatch between expectations and what the Municipality of Amsterdam can deliver within existing planning cycles.

Lack of an integrative tool: inspiration, collaboration, and road-specific guidance

This diagnosis directly motivates the need for a support tool that not only structures technical choices but also offers a shared reference point around which fragmented stakeholders and residents can negotiate GOW30 designs. Across interviews, experts agreed that no existing instrument helps practitioners translate the complex mix of design criteria, stakeholder interests, and contextual constraints into a clear and accessible process. Current handbooks, checklists, and assessment frameworks contain substantial content but lack procedural guidance; as the expert from DTV noted, *"there is a lot of assessment framework, but no process that tells you how to use it."* For this procedural guidance, instead of parallel interpretations of the same problem, a tool servicing public dialogue could allow users to react to a common visualisation or scenario, making trade-offs more transparent and supporting co-creation.

Interviewees also emphasised that practitioners often struggle to visualise what a credible GOW30 layout should look like in a specific road context. While they know which measures exist, such as narrowing lanes or adding humps, the challenge lies in selecting and combining them meaningfully. The municipal traffic designer described this as a lack of *"design inspiration,"* noting that teams can list interventions but still lack a shared picture of what the road could become.

3.3 Co-creation insights

To gain insights on the third sub-research question — *How do expert stakeholders perceive the challenges and opportunities within the current GOW30 decision-making and implementation process?* — a co-creation session was organised. This phase in the Living Lab process served as the final validation step in the iterative development of the decision-making support tool. By engaging directly with experts, the initial technical framework (drafted from the policy review and interviews) was validated with professionals from the relevant field.

3.3.1 Design workshop

Part of the iterative process was the design workshop. Within this workshop, invited stakeholders (as outlined in §2.2.3) were tasked with identifying relevant design criteria, discussing their implications and importance, and applying them to a real-world case. The aim of this is to validate whether the framework's initial logic aligns with professional practice.

The workshop made use of a worksheet in order to gather each stakeholder's individual insights in a structured manner. This sheet, included in Appendix 2, contains three distinct parts. In the first part, stakeholders were asked to list the criteria their organisation focuses on in the redesign process of a GOW30 road. Then for each criterion, they were also instructed to rate its relevance on a scale of 1 (barely relevant) to 4 (highly relevant), and provide a brief explanation for their decision. The second part focused on translating the most important criteria to specific design choices that flow from them. Lastly, in the third part, participants were asked to apply the insights from the first 2 parts on a real-world case and draw a road profile (cross-section) representing what they deem as an ideal layout. Each participant then presented their results and design to the rest of the group.

All completed worksheets were analysed to identify patterns in stakeholder responses. Through a process of descriptive coding, the raw criteria listed by participants were grouped into overarching themes. To make this visually clear, figure 8 presents all criteria mentioned during the design workshop.

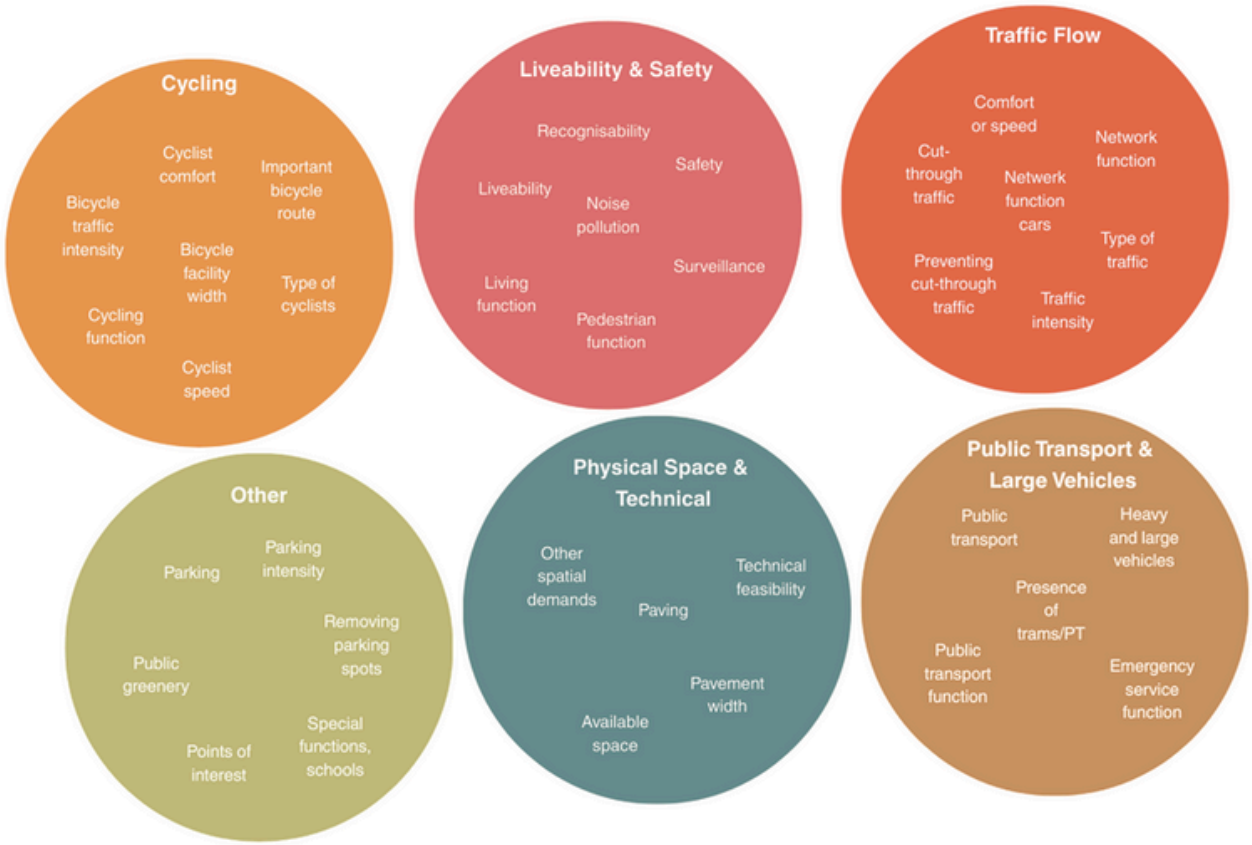


Figure 8: All the criteria mentioned by experts during the design workshop.

The analysis revealed interesting results; the criterion that was mentioned most frequently by stakeholders was public greenery (n=6). After greenery, the most frequently mentioned criteria were parking and traffic intensity (of both motorized and bicycle traffic). When comparing the criteria mentioned by stakeholders with the list of 11 criteria as part of the initial design framework, it becomes apparent that the workshop mostly validated the preliminary list. Five of the eleven items were explicitly mentioned during the workshop: motor vehicle intensity, bicycle traffic intensity, parking intensity, and public transport/emergency routes. Additionally, several criteria in the initial framework also aligned conceptually but were worded slightly different by the stakeholders, such as land use, vulnerable groups, and freight/heavy traffic. Given the frequent mentioning of public greenery, this criterion could be added to the tool in the future.

Based on these findings, the initial list of 11 criteria was largely retained. The only modification made was related to criterion #5 of the initial list — 'Public transport and emergency service routes' — since the participants mentioned trams and emergency services as separate criteria. Additionally, during the panel discussion, a representative from the Municipality of Amsterdam highlighted that the spatial impact of trams differ from those of buses or emergency vehicles. To reflect this comment and the local context of Amsterdam, this single criterion was split into 3 distinct criteria: tram routes, bus routes, and emergency service routes.

3.3.2 Panel discussion

For the first theme (Governance, Design and Institutional Dynamics), the following two statements were voted to be discussed by the participants:

Statement 1: Rule change vs. physical redesign: *"Implementing GOW30 without redesigning the road undermines residents' trust in the policy."*

To start this discussion, several traffic experts acknowledged the ambition of the Amsterdam's large-scale implementation of GOW30 but also raised points of criticism regarding its execution. The Municipality of Amsterdam representative emphasized that the chosen strategy was the right and only option, while traffic experts noted that a slower, small-scale implementation strategy would also have been effective. Ethical questions were raised concerning fining on roads that were not designed for 30 km/h. Police have largely refused to enforce traffic speed fines on high non-compliant roads due to the high manpower required.

The public perception and interpretation of traffic rules and compliance was also discussed next. Some participants noted that they expect rules to be followed based merely on their clear communication: *"We agreed that a white sign with a red border and the number 30 means you cannot drive faster than 30. You know that, right?"* SWOV commented that people may perceive GOW30 as being tolerant towards speeding more than other road types, potentially due to its transitionary phase or because roads still visually resemble 50 km/h roads. Participants emphasized that speed limits must be recognizable in the environment: *"It must be clear from the surroundings what the speed limit is,"* and noted that inconsistent speeds create uncertainty for pedestrians and cyclists. Some argued that if roads are not redesigned to uniform standards and clarity, the policy remains dangerous for vulnerable road users. The concept of self-explaining roads are therefore a major priority in the success to this policy (Geem et al., 2013).

However, when zooming out, some questioned the generally high costs of physical redesign, suggesting that millions should not be required to ensure compliance when the agreed rule is 30 km/h: *"Why do you have to spend millions to redesign the road so that everyone automatically drives 30? We already agreed that the speed is 30."* These perspectives illustrate contrasting views on the power and purpose of rules, with some seeing non-compliance as natural while others consider it entirely unacceptable.

Finally, the discussion addressed the division of responsibilities between municipal and national authorities. The municipality bears all responsibility for implementation, yet rules and funding come from the national government: *"The national government keeps its hand on the purse while municipalities are expected to make enormous investments."*

Statement 2: Central guidelines vs. local adaptation: *"One city, one standard does not work; with our tool we can show, road by road, what GOW30 actually requires."*

The second discussion statement brought forward the tension between the need for recognisability and the realities of Amsterdam's highly varied road environments. Across the panel, participants agreed that a coherent visual identity for GOW30 is essential for user understanding. Recognisability forms the foundation for behavioural compliance and social norming, and a fragmented or inconsistent visual language risks weakening the credibility of the 30 km/h regime (SWOV, 2019). Yet this need for clarity conflicted with the everyday complexity of the network. Many GOW30 roads traverse multiple spatial typologies within only a few hundred metres, transitioning from mixed-use or commercial areas to residential segments. Relating this to Schepers et al. (2025), the interrelation between speed behaviour and environmental characteristics is key consider here. Experts argued that applying one uniform solution across such varied conditions is neither feasible nor desirable, because certain segments require more intensive interventions than others.

Discussions emphasised that differentiation becomes particularly relevant where road function shifts. For instance, segments carrying high bus frequencies or acting as local distribution corridors cannot be redesigned in the same way as roads that do not carry these functions. Several panel members questioned whether long corridors should remain uniformly 30 km/h when contextual changes might justify a return to 50 km/h speed limits.

Others warned, however, that allowing exceptions risks weakening the clarity of the newly established uniformity. Even subtle inconsistencies in layout or signage could lead to confusion among drivers but also cyclists and pedestrians, who experience these environments more continuously and are more vulnerable when speeds fluctuate unexpectedly (Municipality of Amsterdam, 2025).

In general, the discussion showed that the largest challenge is not the existence of variation itself, but the absence of a shared framework for when and how differentiation is justified. Without such a structure, decisions risk becoming subjective, difficult to compare across projects, and misaligned between municipal departments.

For the second theme, 'Behavioural Change, Design and Social Norms', two more topics were discussed: the first in depth, and the second more briefly due to time constraints.

Statement 2: Speed as efficiency vs. slowness as care: *"GOW30 only works when 'driving slowly' feels like part of good urban living, not a waste of time."*

The debate to this statement opened with a fundamental contrast that had also appeared throughout the interviews: the different mentalities the same person can adopt as a driver versus as a resident. One expert summarized it: *"As a road user you want speed; as a resident you want safety."* This duality mirrors the behavioural theory emphasising that compliance is not only a cognitive process, but highly context-dependent and shaped by users' perceived environment (Yadav & Velaga, 2021).

Participants also reflected on the Municipality of Amsterdam's communication strategy surrounding the 2023 mass-transition. Because large-scale redesign was not immediately feasible after this, the municipality relied heavily on campaigns to shift social norms. Several noted that this softer behavioural approach had limited effect, confirming insights from Gargoum et al. (2016) and (CROW, 2023) that physical (re)design remains foundational in achieving behavioural adaptation.

Campaigns alone could *"not override road geometry"*, and compliance improved only marginally without credible environmental cues. Nevertheless, the SWOV-participant observed that acceptance of the 30 km/h policy has grown faster than expected (Municipality of Amsterdam, 2025). It was argued that norm adaptation tends to stabilise over time as people internalise new traffic rhythms. Others agreed that habitual change is emerging, though they warned that this requires persistent clarity in both design and governance.

A broader cultural critique emerged around societal expectations of efficiency. Several participants framed the difficulty of embracing low-speed urban mobility as part of a wider *"culture of acceleration"* that values speed, optimisation, and productivity beyond the context of traffic alone. From this viewpoint, the challenge for GOW30 is not merely redesigning roads but shifting a cultural narrative: slowness as care, safety, and liveability rather than inefficiency. Literature on self-explaining roads reinforces this, arguing that intuitive environments help encode new behavioural expectations (Geem et al., 2013; Qin et al., 2020). Participants noted that without such reinforcement, slowing down is perceived as a burden rather than a civic contribution.

Lastly, some participants raised the concept of policy lock-in. Amsterdam's decision to convert the entire distributor network at once was described as irreversible: *"As a governing body, when you decide everything becomes GOW30, it will not be reversed."* While this accelerates norm-setting, several experts warned that the sheer scale risks embedding suboptimal designs if upgrades lag behind regulation. This concern directly relates to the Municipality of Amsterdam's (2025) finding that recognisability remains inconsistent across corridors, weakening credibility and trust.

Statement 3: Control-oriented vs. guided confusion: *"A little confusion in traffic can be the most powerful safety measure."*

Participants immediately challenged the premise by emphasising that "intended confusion" does not translate well onto GOW30 roads as a strategy to lower speed limits. While some Amsterdam roads, such as shared spaces or low-volume crossings, successfully use interpretive uncertainty to slow drivers, panel members stressed that this approach is inappropriate where traffic volumes, speeds, and vulnerable users are high. This aligns with Distefano & Leonardi's (2023) argument that unpredictability only works under tightly controlled conditions, and with SWOV's warning that ambiguous layouts can undermine perceived safety for older or stress-sensitive users.

Supporting this, representatives from the Fietsersbond strongly emphasised inclusivity: older pedestrians, children, and people with mobility challenges often feel unsafe in environments relying on informal negotiation. They referenced case examples such as Rozengracht and IJ-boulevard, where minimal regulation was meant to induce caution, but instead produced anxiety and avoidance among users. Panel members all agreed this logic cannot be transferred to GOW30 corridors, where consistency, recognisability, and predictability are essential components of self-explaining road design (CROW, 2023; SWOV, 2019).

3.4 Discussion

At the start of this project, the working hypothesis was that existing GOW30 guidelines could be translated into clear categories that would directly structure design decisions. The combined results from the policy review, expert interviews, and co-creation activities demonstrated that this assumption was overly linear. Rather than a purely technical challenge, GOW30 design emerged as a socio-technical process shaped by interpretation, negotiation, and institutional context.

In response, the research first reconstructed GOW30 policy and literature into a structured design framework, which was subsequently stress-tested through interviews and co-creation to uncover where design guidance breaks down in practice. This process revealed that the primary challenges do not lie in identifying design measures, but in navigating decision-making processes and engaging diverse stakeholders around context-specific trade-offs. Consequently, the project shifted towards developing a tool that supports shared understanding and active stakeholder engagement. This section connects the key Living Lab insights to the resulting design logic of the decision-making support tool, addressing the final research sub-question outlined in §1.3.

3.4.1 Living Lab lessons

Synthesizing the Living Lab insights highlights several recurring points where theoretical ambitions diverge from everyday practice. Rather than exposing a lack of awareness among stakeholders, these lessons illustrate shared challenges in translating widely recognised issues into actionable decision-making processes. They therefore inform both broader reflections on design governance and the functional logic of the final tool.

Technical ideals vs. implementation reality

The co-creation session revealed a tension between conceptual design ambitions and the practical constraints of implementation.

While advisory and research-oriented organisations tend to emphasise coherence, recognisability, and long-term design principles, municipal practitioners must account for budget cycles, maintenance planning, and political timing. This does not reflect disagreement over objectives, but rather different operational realities.

Administrative compliance vs. experienced safety

Across interviews and the panel discussion, participants stressed that formal compliance with design standards does not automatically translate into perceived safety or behavioural change. Roads that technically meet GOW30 requirements may still visually communicate higher speeds, undermining credibility and compliance. This insight reinforces the importance of visual feedback in decision-making. By enabling realistic visualisations of design choices, the tool responds to the need for assessing how roads are experienced in practice rather than relying solely on abstract criteria or numerical thresholds.

The narrative of balance vs. reality of silos

While policy frameworks promote a balance between traffic efficiency and safety/ liveability, the Living Lab discussions showed that these goals are often addressed through separate institutional lenses. Stakeholders are generally aware of this fragmentation, yet there is limited shared understanding of how to navigate trade-offs collaboratively. Rather than resolving these tensions, decision-making often depends on which perspective dominates a given project. The tool is therefore positioned not as an immediate solution to institutional fragmentation, but rather to make trade-offs explicit and comparable, supporting more transparent dialogue across departments and stakeholder groups.

3.4.2 Implications for decision-making support tool

Taken together, the Living Lab insights highlight a clear gap in the current landscape. The existing design framework provides several criteria (see §3.1) but no integrative, road-specific tool currently exists that can inspire alternative designs and structure discussion between these fragmented stakeholders.

To address this gap, the complexity of GOW30 design had to be distilled into an operational set of categorizations. As the results highlight that every road demands a bespoke approach, the digital tool requires standardised input to function. The eleven contextual criteria (see §3.1) identified in the policy review represent the most concrete and recurrent factors shaping design decisions. These criteria are a simplification, but a necessary one: they translate a bespoke design logic into a structured, explainable format. All were structurally explored through the design framework and assessed on their significance in a design tool during the co-creation, with only two key critique points emerging: different forms of key road users (buses, trams and emergency services) should be treated separately, and more concrete category formulations should be provided, e.g., instead of simply “high parking demand,” the tool could indicate high demand using a data-led public source as a reference to guide users. At the same time, participants valued the tool’s relative simplicity and its limited reliance on complex datasets, as this supports accessibility and broader public understanding.

In the end, the final tool (as presented in next chapter) reflects a normative choice where we prioritize physical and functional indicators that directly influence the speed-limit compliance. Here, we deliberately excluded factors such as financial budgeting, and external planning cycles like sewer maintenance. While the co-creation panel acknowledged these as critical “dealbreakers” in reality, they were deemed outside the scope of our prototype aimed at facilitating design dialogue rather than full project management.

4. Product

This chapter concludes the Living Lab research and introduces the final GOW30 decision-making support tool. It reflects how the research informed the product's current form and outlines how the tool can be applied, developed further, and scaled in future practice.

4.1 Conclusion

This research followed a Living Lab process that iteratively combined academic theory, policy analysis, and expert knowledge through interviews and co-creation to address the challenges surrounding GOW30 design implementation in Amsterdam. Starting from the main problem identified, namely the misalignment between policy ambitions, road design, and behavioural compliance on newly designated 30 km/h arterial roads, the project moved through successive phases of exploration, prototyping, and analysis. Each step informed the next and jointly shaped both the analytical insights and the development of the final digital product.

The research showed that decision-making on GOW30 redesign is embedded in a complex governance structure. Road redesign proved to be a highly impactful yet infrequent intervention, typically tied to regular maintenance cycles or broader policy agendas rather than speed management alone. Amsterdam's rapid, city-wide implementation of GOW30 successfully created an impulse for social norm change, but it also exposed a major challenge: many roads retained a former 50 km/h appearance, making compliance difficult to achieve and costly to enforce. Designers operated within a layered policy framework, most notably the *Puccini Handboek Rood*, which is influenced by national guidelines (CROW, 2023; Municipality of Amsterdam, 2024). While this framework provided technical direction, it left limited room to systematically explore context-specific trade-offs.

By definition, GOW30 roads functioned as *"grey roads"* that combine conflicting roles of traffic distributors and liveability (CROW, 2021b; SWOV, 2019). This makes design choices inherently complex.

Expert insights revealed tensions around the necessity of self-explaining road principles. Whereas municipal practice often relied on signage and gradual adaptation, road safety research consistently argued for physical redesign to achieve credible speed limits. In addition, the technocratic nature of traffic engineering, characterised by specialised expertise and data-heavy reasoning, made design trade-offs difficult to communicate, limiting inclusive and co-creative approaches with non-expert stakeholders.

The research demonstrated that existing design criteria, expert reasoning, and policy guidelines could be translated into a transparent and structured decision-making framework when technical variables were explicitly linked to governance realities. This framework combined recurring design elements with contextual criteria that shape their applicability. Namely, expert stakeholders perceived both constraints and opportunities in the current GOW30 implementation process. While the research highlighted the tool's potential future use for more technical applications by traffic engineers, stakeholders also emphasised the importance of enforcement, policing, and collective responsibility. At the same time, the underlying logic of the framework was considered transferable to other urban contexts, provided local conditions are accounted for.

Building on this, the framework was translated into an implementation-oriented digital tool that uses AI-modelled visualisations to make the consequences of design choices tangible and accessible. Insights from the Living Lab were iteratively integrated into the tool: all tool aspects were based on policy documents, while the co-creation process validated and refined the contextual criteria and confirmed that its primary use lies in structuring public dialogue. As a result, the tool functions as a shared reference point through which decision-makers can engage not only with consultants and researchers, but also with residents, transport companies, and emergency services.

To then formulate an answer to the central research question — *How can a decision-making support tool contribute to the public dialogue on the governance and implementation of GOW30 design in Amsterdam?* — this study showed that such a tool can play a meaningful role in structuring that dialogue. It does so by translating complex, expert-driven considerations into visual, comparable, and negotiable scenarios that can be understood and discussed by a wider range of stakeholders. The bridge between research and product is therefore explicit: empirical insights into behavioural theory, policy constraints, and stakeholder dynamics directly shape the tool's structure, criteria, and visual logic. The tool is intended to be used by decision-makers, particularly within DRO-DMI, who can consequently better include and co-create with non-expert stakeholders.

While this conclusion has clarified the reasoning behind the development of the decision-making support tool and its intended role in GOW30 governance, it does not yet explain how the tool operates in practice. The Living Lab research established *why* such a tool is needed and what it should enable in terms of dialogue, transparency, and collaboration. The following section therefore shifts from analytical reflection to technical explanation. It showcases how the research insights are translated into practical choices within the product, and explains the underlying structure, logic, and mechanisms of the proof of concept-tool as it currently functions.

4.2 Product description

The GOW30 decision-making support tool is a published outcome of this Living Lab project and can be accessed using this URL: https://digitaltoolsformobility.imagifight.com/.

It is important to note that the current tool is a proof of concept. The current version should be viewed as an early exploration intended for demonstration of how a digital tool can facilitate complex multi stakeholder and interest redesign processes. For future improvements, the underlying framework requires further refinement to yield more comprehensive and accurate results. This could be in the form to further validation of the scoring logic or adding additional contextual criteria and design variables.

Regarding the visual output, the tool first calculates an 'optimal' design based on the user input and subsequent scoring of the chosen inputs. This design gets produced as a textual output, which is then passed along with a static instruction prompt to the OpenAI image API, along with the uploaded image of the original road layout. Because this is a general image generation model rather than a specifically trained model on Dutch road design layouts, it may occasionally generate inaccurate results.

Because the image generation uses a general model rather than one specifically trained on Dutch road design layouts, it may occasionally generate inaccurate results or artefacts. However, the textual output remains accurate, as it is calculated directly by the tool's internal logic. Therefore, users should regard the textual recommendations as the primary source, while the generated image serves as an illustrative reference. These limitations are further addressed in §4.3.

4.2.1 Purpose and target group

The primary goal of the tool is to increase transparency within the decision-making process of GOW30 road redesigns. Because of the inherent dual function of GOW30 roads, the ideal balance of priorities varies per specific road case, which makes it hard to apply a uniform standardised design approach.

To bridge this gap, this tool was developed to bring structure to this process. It is designed to be mainly used by expert stakeholders, namely designers in municipalities or consultants, but is intended to be used during participation and co-creation sessions with non-expert stakeholders. By on-the-spot visualising how specific local contextual criteria influence the road design choices, the tool allows experts to transparently demonstrate to residents why certain design choices are more 'suitable' over others for their specific road.

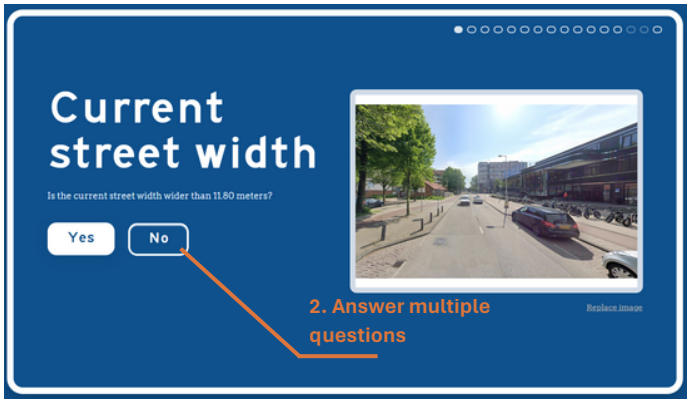
4.2.2 Tool usage

Upon landing on the welcome page that briefly explains the workings and goal of the tool, users are then shown a ‘proof of concept’-disclaimer like described earlier. After clicking on ‘Continue’, they are prompted to upload an image of a road they would like to redesign.



1. Upload an image of your street

It then guides them through a questionnaire that asks them to answer questions related to each specific contextual criterion, and additionally lets them select up to 3 criteria they deem as extra important in this case.



2. Answer multiple questions

After successfully completing this categorization, the tool generates 3 images of a redesigned road, each with a different focus: traffic flow, liveability, and balanced. Users can click on either of the 3 images to see the full redesign, which comes with a textual summary of the ideal design given their input, and why these elements were picked.



3. Review AI-generated redesign

4. Review design recommendations

4.2.3 Visual

In addition, we promoted the product through a foldable flyer explaining GOW30 in Amsterdam, including a concise user guide and a QR code as a low-threshold access point (Appendix 4).



Figure 9: Product user guide and QR code providing access to the product, alongside a visual of the promotional flyer for the website.

4.2.4 Technical architecture

The GOW30 decision-making support tool was developed as a lightweight browser-based single page application utilising standard web technologies: HTML5, CSS3, and ES6 JavaScript. This technical stack was chosen to keep the tool's design simple and lightweight, and avoids the need of a complex backend infrastructure.

Front-end architecture

The tool's structure is defined in a single HTML document, which is then separated into <*section*> elements that each contain a certain part of the tool workflow: the landing page, the questions form, priority selection, and the results display. For styling, the tool relies on a single CSS document that uses CSS variables to maintain a consistent styling, and employs CSS Grid and Flexbox layouts to make the layout automatically adapt to different screen sizes, like on mobile devices.

Tool logic

The tool logic relies on the design framework that was constructed using all insights from the expert interviews and policy review, and validated and improved through the co-creation session with expert stakeholders (as described in the previous chapter).

Within this framework, a total of four design variables are included. Within each variable, there are multiple possible design elements:

1. **Bicycle facilities**
   a. Fully separated bicycle lanes (fietspaden)
   b. On-road bicycle lanes (fietsstroken)
2. **Road width**
   a. Narrow
   b. Medium
   c. Wide
3. **Paving type**
   a. Asphalt
   b. Bricks
4. **Traffic calming measures**
   a. Low
   b. Medium
   c. High

For the purpose of this proof-of-concept, categorical variables were chosen over numerical values to maintain architectural simplicity. While a fully operational traffic engineering tool would ideally use precise numerical inputs to calculate exact cross-sections, this qualitative approach was picked to keep the prototype lightweight. By doing so, it becomes possible demonstrate the core decision-making process of a tool without requiring the complexity of a more complicated calculation logic. This can however be added in the future, which will be elaborated upon in §4.3.

Across these four variables, the framework consists of a total of 10 distinct design elements. The tool determines the optimal design element for each variable based on the 13 contextual criteria:

1. **Road width**, with the binary category whether the road (including kerb space) is wider than 11,8 meters or not. Refer to §3.1.1 for further elaboration on this specific figure.
2. **Land use**, refers to whether the surrounding area has a primary residential or commercial function; or both.
3. **Pedestrian crossing** needs, with the binary category whether a lot of pedestrians need to cross this road.
4. **Vulnerable road** users, refers to whether there are considerable amounts of vulnerable road users close to, or using this road. This is applicable when e.g. schools or daycare centres are nearby.
5. **Tram routes**: does the road contain a tramway? Users can refer to https://maps-vervoerregio.nl/ov/
6. **Bus routes**: is the road part of an active bus route? Users can consult this at https://maps-vervoerregio.nl/ov/
7. **Emergency service routes**: is the road regularly used by emergency services like police, ambulances or fire trucks?
8. **Freight traffic** refers to whether many heavy vehicles use this road regularly. For example, if there are major logistical routes on these roads, proximity to industrial areas, factories, Port of Amsterdam.

9. **Traffic intensity** relates to the approximate number of vehicles per day using this road. This is divided into three subcategories: low (less than 6.000 vehicles), medium (between 6.000 and 10.000 vehicles), and high (over 10.000). These figures are deduced from Goudappel's GOW30 Toolbox (Goudappel, 2025).
10. **Bicycle traffic**: is this road heavily used by cyclists, for example as part of a main cycling route? Users can refer to https://maps.amsterdam.nl/fietsnetten/
11. **Parking demand**, refers to whether the parking demand along this road is high (according to road user/residents' experiences)? Users can consult this data from Maps Amsterdam: https://maps.amsterdam.nl/parkeerdruk/
12. **Pedestrian traffic**: is the kerb space along the road heavily used by pedestrians?
13. **Speed limit compliance**, refers to whether drivers generally comply with the speed limit of 30 km/h.

The selection process of the 'ideal' design element within each variable is driven by a cumulative scoring matrix. For each of the 13 criteria, the user's specific input assigns a predefined weighting rule. A selected answer assigns a score: +1 (favourable) or -1 (unfavourable) to specific design elements within the variables. For example, indicating that a road is a is part of an active bus route assigns a +1 score to "fietspad" (bicycle facilities), while assigning a -1 to "bricks' (paving). This logic is transferred into JavaScript using the 'EFFECTS' constant, which is a multidimensional associative array that maps specific user inputs to numerical weights across the four design variables.

Upon completion of the input questionnaire, the logic aggregates the scores for each design element using a custom written JavaScript function. The design element with the highest cumulative score within each respective variable is selected as the 'winning' element, and then included in the final design recommendation.

Once the calculation is complete, a text string with the highest scoring design elements together with a pre-written static instruction prompt is passed to OpenAI's Image Generation API. The resulting image is then displayed in the user interface. Below this image, the tool provides a summary explaining the design choices. These explanations are directly tied to the hard-coded scoring logic; each criterion/design-element pair consists of a pre-defined written explanation alongside its numerical score (+1 or -1), which allows the tool to provide context on why specific elements were chosen.

4.3 Limitations & future research

This research has provided an extensive analysis on how a digital decision-making support tool could contribute to a more transparent, collaborative, and well-informed dialogue on the implementation of Amsterdam's GOW30 roads. While the Living Lab approach enabled meaningful engagement with expert stakeholders and supported iterative tool development, several methodological and practical limitations shaped both the process and the outcomes. However, these limitations also open opportunities for further refinement of the tool and the wider governance processes it is intended to support in the future.

A first limitation can be found in the range of stakeholders involved. Although co-creation formed the main part of our research focus, several relevant parties remained absent during these activities. For example, emergency services, public transport operators and resident advocacy groups who also play a direct role in how GOW30 designs performs in practice. Their absence means the tool reflects only a partial view of the decision-making process. In particular, trade-offs between accessibility, response, times, road usability and liveability could not be fully explored. Including these groups in future iterations is essential to ensure that the tool can support cross-sector alignment rather than reinforcing existing silos.

A second restriction concerns the tool's function in facilitating dialogue and overcoming institutional barriers. The tool can create and visualize alternative road design scenarios, but its role in resolving political tensions, competing interests, or fragmented responsibilities is indirect. The tool does not replace a process of negotiation or governance, instead it structures the conversation by making assumptions, priorities and trade-offs more explicit. In its current form, the tool supports shared understanding rather than decision-making authority. Future development could further enhance this function by incorporating features that allow stakeholders to compare side-by-side scenarios and trace how changes to criteria weights influence outcomes. Such modifications may provide better support for dialogue between different stakeholders

A third limitation is that the project lacks any aspects in finances. Cost is a decisive factor in road redesigns, but current cost estimates for isolated GOW30 design measures were not considered due to time constraints and variability of (estimated) costs in different sources. This omission means that users cannot yet compare scenarios in terms of financial feasibility, which restricts the practical use of the tool in planning contexts. Integrating financial data in future iterations would strengthen the tool's strategic value.

Another obstacle relates to the weighting system underling the decision support tool's logic. The weights currently assigned to design criteria, such as safety, width of road, or traffic performance, are informed by expert reasoning but not grounded in quantitative measurements or empirically validated models. This makes the output helpful for structuring discussions, but not yet suitable for official evaluation or governance justifications. Future research could build on traffic conflict studies, comfort metrics, or traffic modelling to develop more precise and context-sensitive weights. This would enable the tool to act not only as a conversation starter but also as an analytical support instrument.

The use of AI-generated visualizations introduces other considerations. Visual scenarios make complex design choices more accessible and help non-designers engage in abstract discussions. Yet, AI-generated images also pose several risks. For example, they might oversimplify spatial constraints or unintentionally misrepresent real-life conditions or reflect biases embedded in logic behind our tool. Furthermore, image generation requires high amounts of energy, hence environmental concerns may appear. To address this, future iterations should incorporate human validation steps, clearer explanations of assumptions, and possibly non-AI visualization extensions.

Lastly, this tool has yet to be applied in realistic situations of decision-making. Future research should be directed at piloting the tool on real-life cases, observing its usage within public dialogues or co-creation session between stakeholders and assessing the impact it may have on decision speed, stakeholder alignment, and perceived transparency.

Overall, these limitations do not weaken the relevance of the findings. Many highlight the complexity of governing GOW30 road redesigns and the need for iterative, collaborative design sessions. This project demonstrates that there is a clear institutional demand for tools that help prioritize interventions, clarify trade-offs, and increase transparency. To reach that potential, future research should expand the stakeholder base, add quantitative and financial foundations to the tool, and test the tool in real-life decision-making scenarios. In doing so, this decision support tool could evolve from a conversation starter into useful tool within the governance process and help shift Amsterdam towards higher compliance and more liveable GOW30 roads.

4.4 Acknowledgements

We would like to thank AMS Institute, the DRO-DMI consortium, SWOV, CROW, Fietsersbond, Goudappel, DTV Consultants, and all other experts who contributed their time, expertise, and perspectives throughout the project. Their insights have been invaluable in shaping this study. It is our hope that the resulting academically informed tool and findings will support practitioners, researchers, and policymakers in impactfully strengthening the future design and implementation of GOW30 roads.

In line with the Living Lab Course guidelines on responsible use of artificial intelligence, we also made use of AI-based tools to support aspects of text structuring, language refinement, and exploratory analysis. These tools were used transparently and critically, with all substantive interpretations, decisions, and academic judgements remaining the responsibility of the research team.

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Appendices

Appendix 1 - Expert interview summaries

Interviewee + role	Organisation	Date (time)	Interviewers	Main interview topics
Head of Traffic Design | GOW30 Toolbox	DTV Consultants	1 Oct 2025 (60 min)	Siem Didier	DTV toolboxes for GOW30. The challenge with 'grey roads'. Stakeholder environment: municipality versus consultant.
Lead Researcher & Program Developer	MIT Senseable City Amsterdam | AMS Institute	1 Oct 2025 (45 min)	Siem Didier	MIT research on predicting speed limit compliance using Road View Imagery and AI. Future Curbs Living Lab project.
AI-based researcher	MIT Senseable City Amsterdam | AMS Institute	10 Oct 2025 (50 min)	Didier	Predictive (data-driven) tools in speeding behaviour. Use of AI in digital tools. GOW30 compliance.
Head of Traffic Design | GOW30 Design Toolbox	Goudappel	8 Oct 2025 (70 min)	Siem Didier	GOW30 decision-making framework. Relationship to other stakeholders. Trade-offs in design. The unique strategy of Amsterdam. Predictive tools in speeding behaviour.
Senior Project Manager + Urban Design & Traffic Engineer	Municipality of Amsterdam | Traffic & Public | GOW30 Project Team	9 Oct 2025 (70 min)	Siem Didier Stan	Implementation of speed limit reduction in 2023. GOW30-design implementation and its challenges. Research report on results of GOW30-policy.
Head Proof of Concept-team	DRO-DMI | AMS Institute	13 Oct 2025 (45 min)	Siem Didier Stan	Requirements to Proof of Concepts. Values within DRO-DMI. Continued development of the tool.
Project Manager Mobility	CROW	21 Oct 2025 (60 min)	Siem Didier Stan	Political dynamics behind GOW30. Role of research institutes versus municipalities. Flexibility of policy guidelines.
Researcher on Road Safety	SWOV	22 Oct 2025 (70 min)	Siem Didier	Exploratory research of GOW30 in 2019. Context-dependency in GOW30 design interventions. Road network hierarchy.
Researcher on GOW30 road safety	SWOV	28 Oct 2025 (60 min)	Siem Didier Stan	GOW30 implementation differences across municipalities. Monitoring impact of GOW30-policies. Publication of new GOW30 design toolbox.

Werkblad Herinrichting GOW30

Naam: .....

Organisatie: .....

Functie/expertisegebied: .....

Deel 1: Inventarisatie criteria

Doel:
Inzicht krijgen in hoe belangrijk welke criteria is voor jouw organisatie bij het ontwerpen van een veilige en functionele GOW30-straat.

Instructie:
Schrijf de criteria op die jouw organisatie meeweegt bij het (her)ontwerpen van een GOW30-straat. Geef per criterium aan hoe relevant het is op een schaal van 1 tot 4, waarbij 1 staat voor 'nauwelijks relevant' en 4 voor 'doorslaggevend voor het GOW30-ontwerp', en licht kort toe waarom dit criterium belangrijk is.

Criteria	Relevantie 1 2 3 4	Toelichting
1.	☐☐☐☐	.....
2.	☐☐☐☐	.....
3.	☐☐☐☐	.....
4.	☐☐☐☐	.....
5.	☐☐☐☐	.....
6.	☐☐☐☐	.....
7.	☐☐☐☐	.....
8.	☐☐☐☐	.....
9.	☐☐☐☐	.....
10.	☐☐☐☐	.....
11.	☐☐☐☐	.....
12.	☐☐☐☐	.....
13.	☐☐☐☐	.....

Deel 2: Vertaling naar ontwerpkeuzes

Doel:
Inzicht krijgen in hoe de door jouw organisatie belangrijk geachte criteria zich vertalen naar concrete ontwerpkeuzes.

Instructie:
Kies vier criteria uit Deel 1 die volgens jouw organisatie het meest bepalend zijn voor het ontwerp van een veilige en functionele GOW30-straat. Noteer per criterium welke ontwerpkeuzes hieruit voortvloeien en waarom.

Criteria	Ontwerpkeuze	Toelichting
1.	.....	.....
2.	.....	.....
3.	.....	.....
4.	.....	.....

Deel 3: Toepassing op de casus

Instructie:
Pas nu de opgedane inzichten toe uit deel 1 en 2 toe op de casus Pieter Calandlaan. Teken een wegprofiel (dwarsdoorsnede) of bovenzicht waarin je de gewenste inrichting weergeeft, Licht kort toe hoe jouw ontwerpkeuzes bijdragen aan een geloofwaardig en kwalitatief GOW30-ontwerp.

Appendix 3 - Discussion statements co-creation

Statements in Dutch (original, used in co-creation):

Thema 1: Bestuur, Ontwerp en Institutionele dynamiek

1. **Regelverandering vs. fysieke herinrichting**: “GOW30 invoeren zonder daarbij de straat te herontwerpen ondermijnt het vertrouwen van bewoners in het beleid.”
2. **Centrale richtlijnen vs. lokale aanpassing**: “Eén stad, één norm werkt niet - met onze tool kunnen we per straat laten zien wat GOW30 daar echt nodig heeft.”
3. **AI-tool vs. menselijke expertise**: “AI verhelpt een belangrijke tekortkoming om beleid direct zichtbaar te maken - pas als mensen het kunnen zien, kunnen ze het ook begrijpen.”
4. **Top-down regulering vs. co-gecreëerde verandering**: “GOW30 werkt alleen als bewoners en bottom-up initiatieven mee kunnen bepalen wat een ‘goed ingerichte straat’ betekent.”
5. **Interdepartementale silo’s vs. geïntegreerd eigenaarschap**: “De grootste barrière voor het effectief herinrichten van GOW30 is het bestuurlijke kader met gefragmenteerde afdelingen en uiteenlopende werkwijzen.”
6. **Verantwoording vs. innovatie-enthousiasme**: “Als we alleen handelen wanneer succes gegarandeerd is, zullen innovatieve ideeën rond het inrichten van GOW30-wegen nooit van de grond komen.”

Thema 2: Gedragsverandering, Ontwerp en Normen

1. **Naleving door beboeting vs. ontwerp**: “Je kunt geen gedragsverandering afdwingen met boetes - alleen ontwerp kan het vanzelfsprekend maken.”
2. **Snelheid als efficiëntie vs. traagheid als zorgzaamheid**: “GOW30 werkt pas als ‘langzaam rijden’ voelt als onderdeel van goed stadsleven, niet als tijdverlies.”
3. **Controlegericht vs. gestuurd verwarren**: “Een beetje verwarring in het verkeer kan de krachtigste veiligheidsmaatregel zijn.”
4. **Individuele verantwoordelijkheid vs. collectieve norm**: “GOW30 werkt pas echt als langzamer rijden voelt als socialiteit, niet als gehoorzaamheid.”
5. **Ontwerpprikkels vs. gedragsmatige gewenning**: “GOW30-kenmerken moeten bestand zijn tegen de dreiging van gewenning en normalisering.”
6. **Radicale transformatie vs. geleidelijke aanpassing**: “Snel GOW30 beleid kan verkeer op korte termijn vertragen, maar heeft op lange termijn negatieve gevolgen door de normverandering te ondermijnen.”

Appendix 4 - The visual

